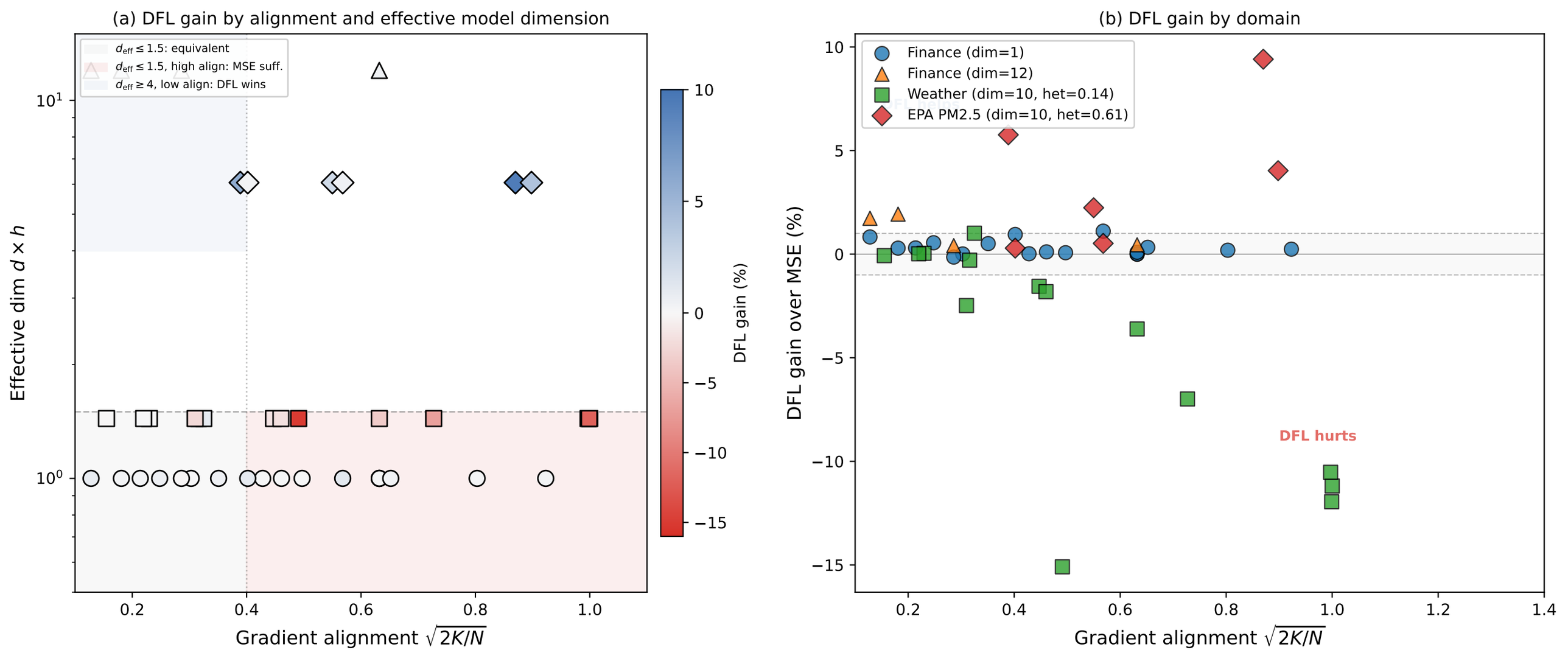
(a) DFL gain by alignment and effective model dimension
$d_{\text{eff}} \leq 1.5$: equivalent
$d_{\text{eff}} \leq 1.5$, high align: MSE suff.
$d_{\text{eff}} \geq 4$, low align: DFL wins
Effective dim $d \times h$
10^1
10^0
0.2
0.4
0.6
0.8
1.0
Gradient alignment $\sqrt{2K/N}$
DFL gain (%)
10
5
0
−5
−10
−15
(b) DFL gain by domain
Finance (dim=1)
Finance (dim=12)
Weather (dim=10, het=0.14)
EPA PM2.5 (dim=10, het=0.61)
DFL gain over MSE (%)
DFL hurts
10
5
0
−5
−10
−15
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Gradient alignment $\sqrt{2K/N}$